Chapter 3
Transport Layer

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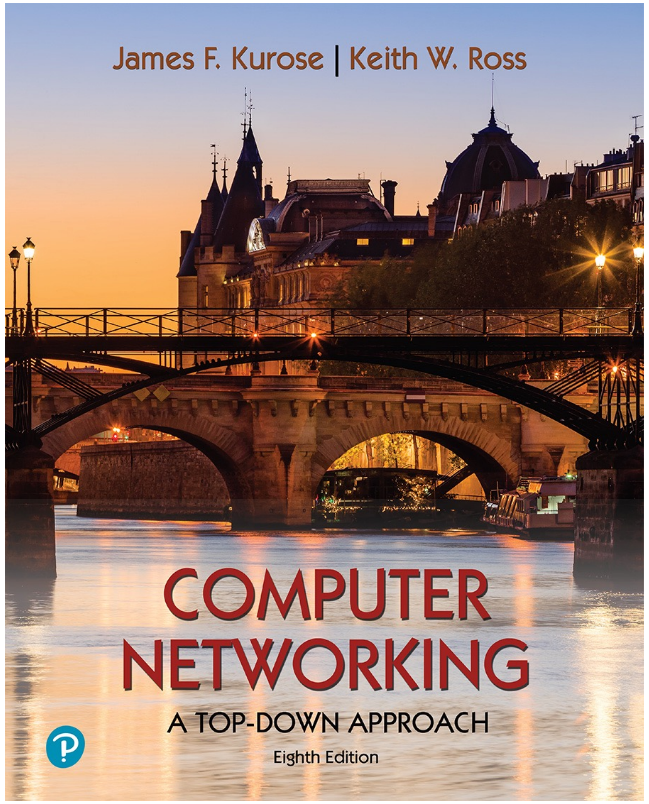
Thanks and enjoy! JFK/KWR



Computer Networking: A Top-Down Approach
8th edition
Jim Kurose, Keith Ross
Pearson, 2020

Question

- Suppose there is a 10 Mbps microwave link between a geostationary satellite (36,000 km)and its base station at earth. Every minute the satellite takes a photo and sends it to the base station. Assume a propagation speed of 2.4×10^8 m/s.
  - What is the propagation delay of the link?
  - What's the bandwidth delay product. $R \times d_{prop}$?
  - Let x denote the size of the photo. What is the maximum value of x for the microwave link to be continuously transmitting?

Question

- Assume you request a webpage consisting of one document and five images. The document size is 1 kbytes and all images are of same size of 50 kbytes. The download rate is 1 Mbps and the RTT is 100 ms. How long does it take over:
  - NP HTTP with serial connection
  - NP HTTP with two parallel connections
  - NP HTTP with six parallel connections
  - P HTTP with one connection

Question

- Consider transferring an enormous file of L bytes from Host A to Host B. Assume an MSS of 536 bytes
  - What is the maximum value of L such that TCP sequence numbers are not exhausted for given MSS?
  - For the L that you obtain in a, find how long it takes to transmit the file. Assume that a s total of 66 bytes of transport, network, data-link headers are added to each segment before the resulting packet is sent over the 155 Mbps link. Ignore flow and congestion control, so segments can be sent back to back.